

Soluções - FME, Volume 06

Capítulo I - Exercícios

1. a) $(6 + 7i)(1 + i) = (6, 7)(1, 1) = (6 - 7, 6 + 7) = -1 + 13i$
b) $(5 + 4i)(1 - i) + (2 + i)i = (5, 4)(1, -1) + (2, 1)(0, 1) = (5 + 4 + 0 - 1, -5 + 4 + 2 - 0) = 8 + i$
c) $(1 + 2i)^2 - (3 + 4i) = (1, 2)^2 - (3, 4) = (1 - 2^2, 4) - (3, 4) = -6$
2. a) $(3 + 2i) + (2 - 5i) = 5 - 3i$
b) $(5 - 2i) - (2 + 8i) = 3 - 10i$
c) $(1 + i) + (1 - i) - 2i = 2 - 2i$
d) $(6 + 7i) - (4 + 2i) + (1 - 10i) = 3 - 5i$
3. a) $(2 - 3i)(1 + 5i) = (2 + 15, 10 - 3) = 17 + 7i$
b) $(1 + 2i)(2 + i) = (2 - 2, 1 + 4) = 5i$
c) $(4 - 3i)(5 - i)(1 + i) = (20 - 3, -4 - 15)(1, 1) = (17 + 19, 17 - 19) = 36 - 2i$
d) $(7 + 2i)(7 - 2i) = (49 + 4, -14 + 14) = 53$
4. $u = 4 + 3i, v = 5 - 2i, uv = ?$
 $uv = (4, 3)(5, -2) = (20 + 6, -8 + 15) = 26 + 7i$
5. a) $(3 + 2i)^2 = (3, 2)^2 = (3^2 - 2^2, 2 \cdot 6) = 5 + 12i$
b) $(5 - i)^2 = (5, -1)^2 = (5^2 - (-1)^2, -10) = 24 - 10i$
c) $(1 + i)^3 = (1, 1)^3 = (1^2 - 1^2, 2)(1, 1) = (0 - 2, 0 + 2) = -2 + 2i$
6. $f(z) = z^2 - z + 1, f(1 - i) = ?$
 $f(1 - i) = (1, -1)^2 - (1, -1) + (1, 0) = (0, -2) + (0, 1) = -i$
7. $f(z) = z^4 + iz^3 - (1 + 2i)z^2 + 3z + 1 + 3i, f(1 + i) = ?$
 $f(1 + i) = (1, 1)^2 = (0, 2)$
 $= (0, 2)^2 + (0, 1)(0, 2)(1, 1) - (1, 2)(0, 2) + (3, 3) + (1, 3)$
 $= (-4, 0) + (-2, 0)(1, 1) - (-4, 2) + (4, 6)$
 $= 2[(-2, 0) + (-1, -1) - (-2, 1) + (2, 3)]$
 $= 2(1, 1) = 2(1 + i)$
8. Prove que $(1 + i)^2 = 2i$ e coloque na forma algébrica $z = \frac{(1 + i)^{80} - (1 + i)^{82}}{i^{96}}$

$$1. (1, 1)^2 = (1 - 1, 1 + 1) = 2i$$

$$2. z = \frac{(2i)^{40} - (2i)^{41}}{(i^2)^{48}} = \frac{2^{40} - 2^{41} \cdot i}{1} = 2^{40} - 2^{41}i$$

$$9. a) i^{76} = (i^2)^{38} = 1$$

$$b) i^{110} = (i^2)^{55} = -1$$

$$c) i^{97} = (i^2)^{48} \cdot i = i$$

$$d) i^{503} = (i^2)^{251} \cdot i = -i$$

10. Prove que $(1 - i)^2 = -2i$ e calcule $(1 - i)^{96} + (1 - i)^{97}$

$$(1, -1)^2 = (1 - (-1)^2, -1 - 1) = -2i$$

$$(1, -1)^{96} + (1, -1)^{97} = (-2i)^{48} + (-2i)^{48}(1, -1) = 2^{48} + 2^{48} - 2^{48}i = 2^{49} - 2^{48}i$$

11. Se $i^2 = -1$, calcule $(1 + i)^{12} - (1 - i)^{12}$

$$= (2i)^6 - (-2i)^6 = -2^6 + 2^6 = 0$$

12. Simplifique $\frac{(2 + i)^{101}(2 - i)^{50}}{(-2 - i)^{100}(i - 2)^{49}}$

$$= \frac{(2, 1)^{101}(2, -1)^{50}}{(-(2, 1))^{100}(-(2, -1))^{49}} = \frac{(2, 1)^{101}(2, -1)^{50}}{(2, 1)^{100}(-1)(2, -1)^{49}} = -(2, 1)(2, -1) = -(4 + 1, -2 + 2) = -5$$

13. A igualdade $(1 + i)^n = (1 - i)^n$ é válida para os números naturais divisíveis por qual natural?

$$\begin{aligned} \bullet \text{ Para } n = 2k, k \in \mathbb{N} &\Rightarrow ((1 + i)^2)^k = ((1 - i)^2)^k \Rightarrow (2i)^k = (-2i)^k \Rightarrow 2^k i^k = (-1)^k 2^k i^k \\ &\Rightarrow (-1)^k = 1 \Rightarrow k = 2k' \\ &\Rightarrow n = 4k'', k'' \in \mathbb{N} \end{aligned}$$

$$\begin{aligned} \bullet \text{ Para } n = 2k + 1, k \in \mathbb{N} &\Rightarrow ((1 + i)^2)^k(1 + i) = ((1 - i)^2)^k(1 - i) \Rightarrow (2i)^k(1 + i) = (-2i)^k(1 - i) \\ &\Rightarrow \frac{(1 + i)(1 + i)}{(1 - i)(1 + i)} = (-1)^k \Rightarrow (-1)^k = i \Rightarrow \perp \end{aligned}$$

$\therefore n$ deve ser divisível por 4.

$$14. a) 2 + 3yi = x + 9i \quad x = 2, 3y = 9 \Rightarrow \begin{cases} x = 2 \\ y = 3 \end{cases}$$

$$b) (x + yi)(3 + 4i) = 7 + 26i \quad (3x - 4y, 4x + 3y) = (7, 26) \Rightarrow \begin{cases} 3x - 4y = 7 \\ 4x + 3y = 26 \end{cases}$$

$$3x - 4y - \frac{3}{4} \cdot 4x + \frac{3}{4} \cdot 3y = 7 - \frac{3}{4} \cdot 26 \Rightarrow -4y + \frac{9}{4}y = 7 - \frac{3 \cdot 26}{4} \Rightarrow -y \left(\frac{4^2 + 3^2}{4} \right) = \frac{28 - 78}{4}$$

$$y = \frac{50}{25} = 2$$

$$\Rightarrow x = \frac{7 + 4 \cdot 2}{3} = \frac{15}{3} = 5 \Rightarrow \begin{cases} x = 5 \\ y = 2 \end{cases}$$

$$\text{c) } (x+yi)^2 = 4i \quad (x^2-y^2, 2xy) = (0, 4) \Rightarrow \begin{cases} x^2 = y^2 \Rightarrow x = y \\ xy = 2 \end{cases} \Rightarrow x^2 = 2 \Rightarrow \begin{cases} x = y = \sqrt{2} \\ \text{ou} \\ x = y = -\sqrt{2} \end{cases}$$

$$15. \text{ a) } 3 + 5ix = y - 15i \Rightarrow \begin{cases} y = 3 \\ 5x = -15 \end{cases} \Rightarrow \begin{cases} x = -3 \\ y = 3 \end{cases}$$

$$\text{b) } (x+yi)(2+3i) = 1+8i \quad (2x-3y, 3x+2y) = (1, 8) \Rightarrow \begin{cases} 2x-3y = 1 \\ 3x+2y = 8 \end{cases}$$

$$\cancel{2x} - 3y - \frac{2}{3}\cancel{3x} - \frac{2}{3} \cdot 2y = 1 - \frac{2}{3} \cdot 8 \Rightarrow y \left(\frac{-9-4}{3} \right) = \frac{3-16}{3} \Rightarrow y = \frac{-13}{-13} = 1$$

$$\Rightarrow x = \frac{1+3}{2} \Rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$$

$$\text{c) } (3+yi) + (x-2i) = 7-5i \Rightarrow \begin{cases} 3+x = 7 \\ y-2 = -5 \end{cases} \Rightarrow \begin{cases} x = 4 \\ y = -3 \end{cases}$$

$$\text{d) } (x+yi)^2 = 2i \Rightarrow (x^2-y^2, 2xy) = (0, 2) \Rightarrow \begin{cases} x^2 = y^2 \\ xy = 1 \end{cases} \Rightarrow \begin{cases} x = 1 \text{ e } y = 1 \\ \text{ou} \\ x = -1 \text{ e } y = -1 \end{cases}$$

$$\text{e) } (2-x+3y) + 2yi = 0 \Rightarrow \begin{cases} 2-x+3y = 0 \\ 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = 0 \end{cases}$$

$$\text{f) } (3-i)(x+yi) = 20 \quad (3x+y, 3y-x) = (20, 0) \Rightarrow \begin{cases} 3x+y = 20 \\ 3y = x \end{cases}$$

$$\Rightarrow 3x + \frac{x}{3} = 20 \Rightarrow 9x + x = 3 \cdot 20 \Rightarrow x = 6 \Rightarrow \begin{cases} x = 6 \\ y = 2 \end{cases}$$

$$16. \begin{cases} x+yi = i \\ xi+y = 2i-1 \end{cases}$$

$$x = i(1-y) \Rightarrow (1-y)i^2 + y = 2i-1 \Rightarrow y-1+y = 2i-1 \Rightarrow y = i$$

$$\Rightarrow x + i^2 = i \Rightarrow x = 1+i \Rightarrow \begin{cases} x = 1+i \\ y = i \end{cases}$$

$$17. (a,b)(c,d) = (ac-bd, ad+bc) = (x,0), x \in \mathbb{R}$$

$$\Rightarrow ad + bc = 0 \Rightarrow ad = -bc \Rightarrow \frac{a}{c} = -\frac{b}{d}$$

$$18. (a, b)^2 \Rightarrow (a^2 - b^2, 2ab)^2 = ((a^2 - b^2)^2 - 4a^2b^2, 4ab(a^2 - b^2))$$

$$\Rightarrow 1. 4ab(a^2 - b^2) = 0 \Rightarrow a = \pm b \ (a \neq 0 \wedge b \neq 0) \text{ ou } a = 0 \text{ ou } b = 0$$

$$\Rightarrow 2. (a^2 - b^2)^2 - 4a^2b^2 < 0 \Rightarrow (a^2 - b^2)^2 < 4a^2b^2$$

$$\bullet a = 0, b \neq 0 \Rightarrow (-b^2)^2 < 0 \Rightarrow b^4 < 0 \Rightarrow \perp$$

$$\bullet a \neq 0, b = 0 \Rightarrow a^4 < 0 \Rightarrow \perp$$

$$\bullet a = \pm b, a \neq 0, b \neq 0 \Rightarrow ((\pm b)^2 - b^2)^2 < 4(\pm b)^2b^2 \Rightarrow 0 < 4b^4, \forall b \neq 0$$

$$\therefore \begin{cases} a = \pm b \\ ab \neq 0 \end{cases}$$

$$19. a) \frac{2}{i} \cdot \frac{-i}{-i} = -2i$$

$$b) \frac{3}{2+i} \cdot \frac{2-i}{2-i} = \frac{6}{5} - \frac{3}{5}i$$

$$c) \frac{1+2i}{3-i} \cdot \frac{3+i}{3+i} = \frac{1}{10} + \frac{7}{10}i$$

$$d) \frac{i^9}{4-3i} \cdot \frac{4+3i}{4+3i} = \frac{i^8 \cdot i(4+3i)}{25} = \frac{-3}{25} + \frac{4}{25}i$$

$$20. a) \frac{1}{i} \cdot \frac{i}{i} = -i$$

$$b) \frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1}{2} - \frac{1}{2}i$$

$$c) \frac{3+4i}{2-i} \cdot \frac{2+i}{2+i} = \frac{2}{3} + \frac{11}{5}i$$

$$d) \frac{1+i}{1-i} \cdot \frac{1+i}{1+i} = \frac{2i}{2} = i$$

$$e) \frac{i^{11} + 2i^{13}}{i^{18} - i^{37}} = \frac{i^{11}(1 + 2i^2)}{i^{18}(1 - i^{19})} = \frac{-1}{i^4 i^2 i(1 - i^{16} i^2 i)} = \frac{1}{i(1+i)} = \frac{1}{-1+i} \cdot \frac{-1+i}{-1+i} = -\frac{1}{2} - \frac{1}{2}i$$

$$f) \frac{1-3i}{3-i} \cdot \frac{3+i}{3+i} = \frac{3}{5} - \frac{4}{5}i$$

$$g) \frac{i^3 - i^2 + i^{17} - i^{35}}{i^{16} - i^{13} + i^{30}} = \frac{i^2(i - 1 + i^{12}i^3 - i^{32}i)}{i^{13}(i^3 - 1 + i^{17})} = \frac{i - 1 - i - i}{i^3(-i - 1 + i^{16}i)} = \frac{-(1+i)}{-i(-1)} \cdot \frac{-i}{-i} = -1 + i$$

$$h) \frac{1+i}{(1-i)^2} = \frac{1+i}{-2i} \cdot \frac{2i}{2i} = \frac{-2+2i}{4} = -\frac{1}{2} + \frac{1}{2}i$$

$$21. \frac{1+3i}{2-i} \cdot \frac{2+i}{2+i} = -\frac{1}{5} - \frac{7}{5}i$$

$$22. \frac{1+i}{i} \cdot \frac{-i}{-i} = 1-i \implies 1+i$$

$$23. z = \frac{1-i}{1+i} \cdot \frac{-2i}{2} = -i \implies z^{-1} = \frac{1}{-i} \cdot \frac{i}{i} = i \implies \overline{z^{-1}} = -i$$

$$24. u^2 - v^2 = 6, \bar{u} + \bar{v} = 1-i, u-v=?$$

$$(u-v)(u+v) = 6 \implies u-v = \frac{6}{u+v}$$

$$\bar{u} + \bar{v} = 1-i \implies \overline{u+v} = \overline{1-i} \implies u+v = 1+i$$

$$\therefore u-v = \frac{6}{u+v} = \frac{6}{1+i} \cdot \frac{1-i}{1-i} = 3-3i$$

25. Provavelmente tem uma informação errada na questão $u = 1$

$$26. \left(\frac{1+i}{1-i} \right)^n, n \in \mathbb{Z}$$

$$\left(\frac{1+i}{1-i} \cdot \frac{1+i}{1+i} \right)^n = \left(\frac{2i}{2} \right)^n = i^n \therefore 1, i, -1, -i$$

$$27. \frac{a+bi}{c+di}, c+di \neq 0$$

$$\frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac-bd+i(ad+bc)}{c^2+d^2}$$

$$\implies ax-bd=0$$

$$\implies ad+bc=0$$

$$28. u = x + yi, v = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \operatorname{Re}(v\bar{u}) = ?$$

$$v\bar{u} = (x + yi)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = \left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x + \frac{y}{2}\right)$$

$$\operatorname{Re}(v\bar{u}) = \frac{x}{2} - \frac{\sqrt{3}}{2}y$$

$$29. z = x + iy, x^2 + y^2 \neq 0, \operatorname{Im}\left(z + \frac{1}{z}\right) = 0, z = ?$$

$$z + \frac{1}{z} = \frac{(x+iy)(x+iy)+1}{x+iy} \cdot \frac{(x-iy)}{(x-iy)} = \frac{(x-iy)(x^2-y^2+2xyi+1)}{x^2+y^2}$$

$$= \frac{(x, -y)}{x^2+y^2}(x^2-y^2+1, 2xy) = \frac{(x(x^2-y^2+1)+2xy^2, 2x^2y-y(x^2-y^2+1),)}{x^2+y^2}$$

$$\implies \operatorname{Im}\left(z + \frac{1}{z}\right) = 2x^2y - y(x^2 - y^2 + 1) = 0 \implies 2x^2y = y(x^2 - y^2 + 1)$$

$$\implies 2x^2 = x^2 - y^2 + 1 \implies x^2 + y^2 = 1$$

30. $z_1, z_2 \in \mathbb{C}, \operatorname{Im}(z_1 + z_2) = \operatorname{Im}(z_1 z_2) = 0$

$$z_1 = a + bi, z_2 = c + di \implies \begin{cases} z_1 z_2 = (ac - bd, ad + bc) \\ z_1 + z_2 = (a + c, b + d) \end{cases} \implies \begin{cases} ad + bc = 0 \\ b + d = 0 \end{cases}$$

$$\implies \begin{cases} ad = -bc \\ b = -d \end{cases} \implies \begin{cases} a = c \\ b = -d \end{cases} \implies z_1 = \bar{z}_2 \text{ ou } z_1, z_2 \in \mathbb{R}$$

31. $z = \frac{2 - xi}{1 + 2xi}, x \in \mathbb{R}, \operatorname{Re}(z) = 0$

$$z = \frac{2 - xi}{1 + 2xi} \cdot \frac{1 - 2xi}{1 - 2xi} = \frac{(2x - 2x^2) + (-4x - x)i}{1 + 4x^2}$$

$$z = \frac{2(1 - x^2) - 3xi}{4x^2 + 1} \implies \frac{2(1 - x^2)}{4x^2 + 1} = 0 \implies 1 - x^2 = 0 \implies x = \pm 1$$

32. $z = \frac{1 + 2i}{2 + ai}, a = ?$

$$\frac{1 + 2i}{2 + ai} \cdot \frac{2 - ai}{2 - ai} = \frac{(2 + 2a) + (-a + 4)i}{a^2 + 4} \implies \frac{-a + 4}{a^2 + 4} = 0 \implies a = 4$$

33. $\operatorname{Im}(z(5 + 8i)) = 0, \operatorname{Re}\left(\frac{z}{1 + i}\right) = 0, z = ?$

$$(x + yi)(5 + 8i) = (5x - 8y) + (8x + 5y)i \implies 8x + 5y = 0$$

$$\frac{x + yi}{1 + i} \cdot \frac{1 - i}{1 - i} = \frac{(x + y) + (-x + y)i}{2} \implies \frac{x + y}{2} = 0 \implies x + y = 0$$

$$\implies \perp$$

34. $\frac{z}{1 - i} + \frac{z - 1}{1 + i} = \frac{5}{2} + \frac{5}{2}i, z = ?$

$$\frac{z(1 + i) + (z - 1)(1 - i)}{(1 - i)(1 + i)} = \frac{5}{2}(1 + i) \implies z + \cancel{z} + z - \cancel{z} - 1 + i = \frac{5}{2} \cdot 2(1 + i)$$

$$2z - 1 + i = 5(1 + i) \implies 2z = 5 + 5i + 1 - i = 6 + 4i \implies z = 3 + 2i$$

35. $z \in \mathbb{C}, \bar{z} = -2zi$

$$z = x + yi \implies x - yi = -2i(x + yi) = -2xi + 2y \implies \begin{cases} x = 2y \\ y = 2x \end{cases} \implies z = 0$$

36. $z = x + yi, u = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \operatorname{Re}(u\bar{z}) = ?, \operatorname{Im}(u\bar{z}) = ?$

$$u\bar{z} = \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)(x - yi) = \left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{-y}{2} - \frac{\sqrt{3}}{2}x\right)$$

$$\begin{cases} \operatorname{Re}(u\bar{z}) = \frac{x}{2} - \frac{\sqrt{3}}{2}y \\ \operatorname{Im}(u\bar{z}) = \frac{-\sqrt{3}}{2}x - \frac{y}{2} \end{cases}$$

37. Prove que $\overline{z^n} = (\bar{z})^n, \forall n \in \mathbb{N}$.

$$z = a + bi \implies z^n = (a + bi)^n = \sum_k^n \binom{n}{k} a^k (bi)^{n-k}$$

$$\implies \overline{z^n} = \sum_k^n \binom{n}{k} a^k (-bi)^{n-k} = (a + (-bi))^n = (\bar{z})^n$$

38. Prove que se a equação $x^2 + (a + bi)x + (c + di) = 0$, com $a, b, c, d \in \mathbb{R}$, admite uma raiz real então $abd = d^2 + b^2c$.

$$\exists x \in \implies x^2 + ax + bxi + c + di = 0 \implies (x^2 + ax + c, xb + d) = 0 \implies \begin{cases} x^2 + ax + c = 0 \\ xb + d = 0 \end{cases}$$

$$\implies x = \frac{-d}{b} \implies \frac{d^2}{b^2} + \frac{a(-d)}{b} + c = 0 \implies d^2 - abd + cb^2 = 0 \implies abd = d^2 + cb^2$$

39. $z\bar{z} + (z - \bar{z}) = 13 + 6i, z = ?$

$$x^2 + y^2 + x + yi - x + yi = 13 + 6i \implies x^2 + y^2 + 2yi = 13 + 6i \implies \begin{cases} x^2 + y^2 = 13 \\ 2y = 6 \end{cases}$$

$$\implies x^2 = 13 - 9 = 4 \implies \begin{cases} x = \pm 2 \\ y = 3 \end{cases}$$

40. $z \in \mathbb{C} | z^3 = \bar{z}$

$$(x^2 - y^2, 2xy)(x, y) = (x^2 - xy^2 - 2xy^2, x^2y - y^3 + 2x^2y) = (x, -y)$$

$$\implies (x^3 - 3xy^2, y^3 - 3x^2y) = (x, y) \implies \begin{cases} x(x^2 - 3y^2 - 1) = 0 \\ y(y^2 - 3x^2 - 1) = 0 \end{cases}$$

$$\implies 1. x = y = 0 \implies z = 0$$

$$\implies 2. x \neq 0 \wedge y = 0 \implies x^2 - 3y^2 - 1 = 0 \implies x = \pm 1 \implies z = \pm 1$$

$$\implies 3. x = 0 \wedge y \neq 0 \implies x^2 - 3y^2 - 1 = 0 \implies y = \pm 1 \implies z = \pm i$$

$$\implies 4. x \neq 0 \wedge y \neq 0 \implies \begin{cases} x^2 - 3y^2 - 1 = 0 \\ y^2 - 3x^2 - 1 = 0 \end{cases} \implies y^2 - 3(1 + 3y^2) - 1 = 0$$

$$\implies -8y^2 = 4 \implies y \in \mathbb{C} \implies \perp$$

41. $z \in \mathbb{R} | z^2 = i$

$$(x^2 - y^2, 2xy) = (0, 1) \implies \begin{cases} x^2 = y^2 \\ 2xy = 1 \end{cases} \implies \begin{cases} x = \pm y \\ xy = \frac{1}{2} \end{cases}$$

$$\implies \begin{cases} x = y \implies x^2 = \frac{1}{2} \implies x = y = \pm \frac{\sqrt{2}}{2} \\ x = -y \implies x^2 = -\frac{1}{2} \implies x \in \mathbb{C} \implies \perp \end{cases}$$

42. $z \in \mathbb{C} | z^2 = 1 + i\sqrt{3}$

$$(x^2 - y^2, 2xy) = (1, \sqrt{3}) \implies \begin{cases} x^2 = 1 + y^2 \\ 2xy = \sqrt{3} \end{cases} \implies 4x^2y^2 = 3 \implies 4x^2y^2 = 4y^2(1 + y^2)$$

$$\implies y^4 + y^2 - \frac{3}{4} = 0 \implies k^2 + k - \frac{3}{4} = 0 \quad (k = y^2) \implies k = \frac{-1 \pm \sqrt{1 - 4 \cdot \frac{3}{4}}}{2} = \begin{cases} \frac{1}{2} \\ -\frac{3}{2} \end{cases}$$

$$\implies y = \begin{cases} \pm \frac{\sqrt{2}}{2} \implies \begin{cases} y = \frac{\sqrt{2}}{2} \implies x = \frac{\sqrt{6}}{2} \\ y = -\frac{\sqrt{2}}{2} \implies x = -\frac{\sqrt{6}}{2} \end{cases} \\ \pm \sqrt{\frac{-3}{2}} \implies y \in \mathbb{C} \implies \perp \end{cases}$$

$$\therefore z = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i \text{ ou } z = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i$$

43. Prove que $x^2 + y^2 = 1 \implies \frac{1 + x + iy}{1 + x - yi} = x + iy$.

$$\frac{1 + x + iy}{1 + x - iy} \cdot \frac{1 + x + iy}{1 + x + iy} = \frac{(1 + x, y)(1 + x, y)}{(1 + x, -y)(1 + x, y)} = \frac{(x^2 + 2x + 1 - y^2, 2y(x + 1))}{x^2 + 2x + 1 + y^2}$$

$$= \frac{(x^2 + 2x + x^2, 2y(x + 1))}{2x + 2} = \frac{2x(x + 1), 2y(x + 1)}{2(x + 1)} = x + iy$$

44. Prove que $\frac{1 + \sin x + i \cos x}{1 - \sin x - i \cos x} = (\tan x + \sec x)i, \forall x \neq \frac{\pi}{2} + k\pi$.

$$\frac{\frac{1}{\cos x} + \frac{\sin x}{\cos x} + i \frac{\cos x}{\cos x}}{\frac{1}{\cos x} - \frac{\sin x}{\cos x} - i \frac{\cos x}{\cos x}} = \frac{\sec x + \tan x + i}{\sec x - \tan x - i} \cdot \frac{\sec x - \tan x + i}{\sec x - \tan x + i}$$

$$= \frac{(\sec^2 x - \tan^2 x - 1, \sec x + \tan x + \sec x - \tan x)}{\sec^2 - 2 \sec x \tan x + \tan^2 x + 1} = \frac{(0, 2 \sec x)}{2 \sec x (\sec x - \tan x)} \cdot \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$= (\tan x + \sec x)i$$

45. a) 4

$$\begin{cases} \rho = \sqrt{4^2} = 4 \\ \cos \theta = 1 \\ \sin \theta = 0 \end{cases} \implies 4 = 4(\cos 0 + i \sin 0)$$

b) $1 + i\sqrt{3}$

$$\begin{cases} \rho = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2 \\ \cos \theta = \frac{1}{2} \\ \sin \theta = \frac{\sqrt{3}}{2} \end{cases} \implies 1 + i\sqrt{3} = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

c) $3i$

$$\begin{cases} \rho = \sqrt{3^2} = 3 \\ \cos \theta = 0 \\ \sin \theta = \frac{3}{3} = 1 \end{cases} \implies 3i = 3(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$$

d) $\sqrt{2} + i\sqrt{2}$

$$\begin{cases} \rho = (\sqrt{2})^2 + (\sqrt{2})^2 = \sqrt{4} = 2 \\ \cos \theta = -\frac{\sqrt{2}}{2} \\ \sin \theta = \frac{\sqrt{2}}{2} \end{cases} \implies \sqrt{2} + i\sqrt{2} = 2(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$$

e) -5

$$\begin{cases} \rho = \sqrt{5^2} = 5 \\ \cos \theta = 1 \\ \sin \theta = 0 \end{cases} \implies -5 = 5(\cos \pi + i \sin \pi)$$

f) $-2i$

$$\begin{cases} \rho = \sqrt{2^2} = 2 \\ \cos \theta = 0 \\ \sin \theta = \frac{-2}{2} = -1 \end{cases} \implies -2i = 2(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})$$

g) $-5 - 5i$

$$\begin{cases} \rho = \sqrt{5^2 + 5^2} = \sqrt{2 \cdot 5^2} = 5\sqrt{2} \\ \cos \theta = \frac{-5}{5\sqrt{2}} = -\frac{\sqrt{2}}{2} \\ \sin \theta = \frac{-5}{5\sqrt{2}} = -\frac{\sqrt{2}}{2} \end{cases} \implies -5 - 5i = 5\sqrt{2}(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4})$$

h) $2 - 2i$

$$\begin{cases} \rho = \sqrt{2^2 + 2^2} = \sqrt{2^3} = 2\sqrt{2} \\ \cos \theta = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \sin \theta = -\frac{2}{2\sqrt{2}} = -\frac{\sqrt{2}}{2} \end{cases} \implies 2 - 2i = 2\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$$

46. a) $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

b) $|\sqrt{2} - i\sqrt{2}| = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{2 + 2} = \sqrt{4} = 2$

c) $|12 + 5i| = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$

d) $|\cos \theta + i \sin \theta| = \sqrt{\cos^2 \theta + \sin^2 \theta} = \sqrt{1} = 1$

e) $|\tan \theta + i| = \sqrt{\tan^2 \theta + 1^2} = \sqrt{\sec^2 \theta} = |\sec \theta|$

f) $|24 + 7i| = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$

47. $\left| \frac{a + bi}{a - bi} \right| = \left| \frac{a + bi}{a - bi} \cdot \frac{a + bi}{a + bi} \right| = \frac{|a^2 - b^2 + 2abi|}{a^2 + b^2} = \frac{\sqrt{(a^2 - b^2)^2 + (2ab)^2}}{a^2 + b^2}$
 $= \frac{\sqrt{a^4 - 2a^2b^2 + b^4 + 4a^2b^2}}{a^2 + b^2} = \frac{\sqrt{(a^2)^2 + 2a^2 + b^2 + (b^2)^2}}{a^2 + b^2} = \frac{\sqrt{(a^2 + b^2)^2}}{a^2 + b^2} = \frac{a^2 + b^2}{a^2 + b^2} = 1$

48. $\left| \frac{1}{1 + i \tan x} \right|, x \neq \frac{\pi}{2} + k\pi$
 $\left| \frac{1}{1 + i \tan x} \cdot \frac{1 - i \tan x}{1 - i \tan x} \right| = \left| \frac{1 - i \tan x}{1^2 + \tan^2 x} \right| = \frac{|1 - i \tan x|}{\sec^2 x}$
 $= \frac{\sqrt{1^2 + \tan^2 x}}{\sec^2 x} = \frac{\sqrt{\sec^2 x}}{\sec^2 x} = \frac{1}{\sec x} = \cos x$

49. a) $3 + 3i$

$$\begin{cases} \rho = \sqrt{3^2 + 3^2} = \sqrt{2 \cdot 3^2} = 3\sqrt{2} \\ \cos \theta = \frac{3}{3\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \sin \theta = \frac{3}{3\sqrt{2}} = \frac{\sqrt{2}}{2} \end{cases} \implies 3 + 3i = 3\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$$

b) $5 - i5\sqrt{3}$

$$\begin{cases} \rho = \sqrt{5^2 + 5^2 \sqrt{3}^2} = \sqrt{25 + 3 \cdot 25} = \sqrt{100} = 10 \\ \cos \theta = \frac{5\sqrt{3}}{10} = \frac{\sqrt{3}}{2} \\ \sin \theta = \frac{5}{10} = \frac{1}{2} \end{cases} \implies 5 - i5\sqrt{3} = 10\left(\cos \frac{\pi}{4} + i \sin \frac{5\pi}{3}\right)$$

c) $-8 - 8i = -8(1 + i)$

$$\begin{cases} \rho = \sqrt{1^2 + 1^2} = \sqrt{2} \\ \cos \theta = -\frac{\sqrt{2}}{2} \\ \sin \theta = -\frac{\sqrt{2}}{2} \end{cases} \implies -8 - 8i = 8\sqrt{2}(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4})$$

d) 11

$$\begin{cases} \rho = \sqrt{11^2} = 11 \\ \cos \theta = \frac{11}{11} = 1 \\ \sin \theta = \frac{0}{11} = 0 \end{cases} \implies 11 = 11(\cos 0 + i \sin 0)$$

e) $2i$

$$\begin{cases} \rho = \sqrt{2^2} = 2 \\ \cos \theta = \frac{0}{2} = 0 \\ \sin \theta = \frac{2}{2} = 1 \end{cases} \implies 2i = 2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$$

f) $i^3 = -i$

$$\begin{cases} \rho = \sqrt{1^2} = 1 \\ \cos \theta = \frac{0}{1} \\ \sin \theta = \frac{-1}{1} = -1 \end{cases} \implies i^3 = 1(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})$$

g) $-\sqrt{3} + i$

$$\begin{cases} \rho = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2 \\ \cos \theta = -\frac{\sqrt{3}}{2} \\ \sin \theta = \frac{\sqrt{2}}{2} \end{cases} \implies -\sqrt{3} + i = 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$$

h) $i(1 + i) = -1 + i$

$$\begin{cases} \rho = \sqrt{1^2 + 1^2} = \sqrt{2} \\ \cos \theta = -\frac{\sqrt{2}}{2} \\ \sin \theta = \frac{\sqrt{2}}{2} \end{cases} \implies i(1 + i) = \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$$

i) $2i(1 - i) = 2 + 2i$

$$\begin{cases} \rho = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2} \\ \cos \theta = \frac{\sqrt{2}}{2} \\ \sin \theta = \frac{\sqrt{2}}{2} \end{cases} \implies 2i(1 - i) = 2\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$$

50. a) $3(\cos \pi + i \sin \pi)$

$$\begin{cases} x = 3 \cos \pi = -3 \\ y = 3 \sin \pi \end{cases} \implies -3$$

b) $2(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

$$\begin{cases} x = 2 \cos \frac{\pi}{4} = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \\ y = 2 \sin \frac{\pi}{4} = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \end{cases} \implies \sqrt{2} + i\sqrt{2}$$

c) $4(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6})$

$$\begin{cases} x = 4 \cos \frac{11\pi}{6} = 4 \cos(2\pi - \frac{\pi}{6}) = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3} \\ y = 4 \sin \frac{11\pi}{6} = 4 \sin(2\pi - \frac{\pi}{6}) = -4 \cdot \frac{1}{2} = -2 \end{cases} \implies 2\sqrt{3} - 2i$$

d) $4(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2})$

$$\begin{cases} x = 5 \cdot 0 = 0 \\ y = 5 \cdot -1 = -5 \end{cases} \implies -5i$$

51. a) $|(1-i)(2-2i)| = |2(1-i)^2| = |2 \cdot 2i| = |4i| = 4$

b) $|(1+i\sqrt{3})^6| = (|1+i\sqrt{3}|)^6 = (\sqrt{1^2 + (\sqrt{3})^2})^6 = 2^6 = 64$

c) $\left| \frac{3+3i}{1+2i} \right| = \frac{|3+3i|}{|1+2i|} = \frac{\sqrt{3^2+3^2}}{\sqrt{1^2+2^2}} = \frac{3\sqrt{2}}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{10}}{5}$

52. a) $|1+i||3| = 3\sqrt{1^2+1^2} = 3\sqrt{2}$

b) $|1-i|^4 = |\sqrt{2}|^4 = 2^2 = 4$

c) $|5+12i||i| = \sqrt{5^2+12^2} = \sqrt{169} = 13$

d) $|1+i||2+2i||4+4i| = \sqrt{2} \cdot \sqrt{8} \cdot \sqrt{32} = \sqrt{2 \cdot 2^3 \cdot 2^5} = \sqrt{2^8 \cdot 2} = 2^4\sqrt{2} = 16\sqrt{2}$

e) $\frac{|1+i|}{|2-2i|} = \frac{\sqrt{2}}{\sqrt{8}} \cdot \frac{\sqrt{8}}{\sqrt{8}} = \frac{\sqrt{2^4}}{8} = \frac{4}{8} = \frac{1}{2}$

f) $\frac{|5i|}{|3+4i|} = \frac{5}{\sqrt{3^2+4^2}} = \frac{5}{\sqrt{25}} = 1$

53. a) $\left(\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^2 = \frac{1}{2}(1+i\sqrt{3})^2 = \frac{1}{4}(-2+2i\sqrt{3}) = \frac{-1}{2} + \frac{\sqrt{3}}{2}$

$$\begin{cases} \rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1 \\ \cos \theta = \frac{-1}{2} \\ \sin \theta = \frac{\sqrt{3}}{2} \end{cases} \implies 1\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$$

b) $\frac{1+i}{i} = 1 - i$

$$\begin{cases} \rho = \sqrt{1^2 + 1^2} = \sqrt{2} \\ \cos \theta = \frac{\sqrt{2}}{2} \\ \sin \theta = \frac{-\sqrt{2}}{2} \end{cases} \implies \sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$$

c) $5(\cos 30^\circ - i \sin 30^\circ) = 5\left(\cos \frac{\pi}{6} + i \sin \frac{-\pi}{6}\right)$

$$\implies 5\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$$

54. $\frac{1}{1-i} - \frac{1}{i} = \frac{i-1+i}{1+i} \cdot \frac{1-i}{1-i} = \frac{1+3i}{2}$

$$\implies \frac{1}{2} + \frac{3}{2}i$$

$$\begin{cases} \rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{9}{4}} = \sqrt{\frac{10}{4}} = \frac{\sqrt{10}}{2} \\ \cos \theta = \frac{\frac{1}{2}}{\frac{\sqrt{10}}{2}} = \frac{\sqrt{10}}{10} \\ \sin \theta = \frac{\frac{3}{2}}{\frac{\sqrt{10}}{2}} = \frac{3\sqrt{10}}{10} \end{cases} \implies \tan \theta = \frac{3\sqrt{10}}{10} \cdot \frac{10}{\sqrt{10}} = 3$$

$$\implies \frac{\sqrt{10}}{2}(\cos(\operatorname{atan} 3) + i \sin(\operatorname{atan} 3))$$

55. $\frac{1}{1+i\sqrt{3}} \cdot \frac{1-\sqrt{3}}{1-\sqrt{3}} = \frac{1-\sqrt{3}}{1+(\sqrt{3})^2} = \frac{1}{4} - \frac{\sqrt{3}}{4}i$

$$\begin{cases} \rho = \sqrt{\frac{1}{4^2} + \frac{(\sqrt{3})^2}{4^2}} = \sqrt{\frac{1}{16} + \frac{3}{16}} = \sqrt{\frac{1}{4}} = \frac{1}{2} \\ \cos \theta = \frac{1}{2} \\ \sin \theta = \frac{-\sqrt{3}}{2} \end{cases} \implies \frac{1}{2}\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$$

56. a) $\frac{5+5i}{2-2i} = \frac{5}{2} \left(\frac{1+i}{1-i}\right) \left(\frac{1+i}{1+i}\right) = \frac{10i}{2 \cdot 2} = \frac{5}{2}i$

$$\begin{cases} \rho = \frac{5}{4} \\ \cos \theta = 0 \\ \sin \theta = 1 \end{cases} \implies \frac{5}{2} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

$$\text{b) } \frac{1}{i} + (2 - i) = \frac{2(1 + i)}{i} \cdot \frac{-i}{-i} = 2(1 - i)$$

$$\begin{cases} \rho = \sqrt{2^2(1^2 + 1^2)} = 2\sqrt{2} \\ \cos \theta = \frac{\sqrt{2}}{2} \\ \sin \theta = \frac{-\sqrt{2}}{2} \end{cases} \implies 2\sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$$

$$\text{c) } \frac{(1 + i\sqrt{3})^3}{i^5} = \frac{(1 + i\sqrt{3})^2(1 + i\sqrt{3})}{i} \cdot \frac{-i}{-i} = (-2, 2\sqrt{3})(1, \sqrt{3})(-i) = 8i$$

$$\begin{cases} \rho = 8 \\ \cos \theta = 0 \\ \sin \theta = 1 \end{cases} \implies 8 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

$$57. \ z, w \in \mathbb{C}, |z| = |w| = 1, 1 + zw \neq 0, \frac{z + w}{1 + zw}?$$

$$\frac{z + w}{1 + zw} \cdot \frac{1 + \overline{zw}}{1 + \overline{zw}} = \frac{z + z\overline{zw} + w + z\overline{w}w}{1 + \overline{zw} + zw + z\overline{z}w\overline{w}} = \frac{z + \overline{w} + w + \overline{z}}{1 + \overline{zw} + zw + 1} = \frac{\operatorname{Re}(z) + \operatorname{Re}(w)}{1 + \operatorname{Re}(zw)}$$

$$\therefore \frac{z + w}{1 + zw} \in \mathbb{R}$$

$$58. \ z_1 = \rho(\cos \phi + i \sin \phi), z_2 = \rho(\sin \phi + i \cos \phi), z_1 - iz_2 \in \mathbb{R}, \mathbb{C}, \mathbb{I}?$$

$$\begin{aligned} z_1 - iz_2 &= \rho(\cos \phi + i \sin \phi) - i\rho(\sin \phi + i \cos \phi) = \rho(\cos + i \sin \phi - i \sin \phi - i^2 \cos \phi) \\ &= 2\rho \cos \phi \implies z_1 - iz_2 \in \mathbb{R} \end{aligned}$$

$$59. \ |z_1| = |z_2| = 1, z_1 + z_2 = 1$$

...